

Donghyun Park

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Citizenship: South Korea

Research interests

E-variable hypothesis testing, Information geometry and its application to statistics

Education

- Mar 2021 – Present **Seoul National University**, *Overall GPA: 4.15*
BA in Statistics, *GPA: 4.30*
Double major in Mathematical Sciences, *GPA: 4.30*
Double major in Computer Science and Engineering, *GPA: 3.90*
- Jul 2023 – Apr 2025 Military service at Republic of Korea Air Force

Honors and scholarships

- Dean's List, (SNU College of Natural Sciences)
Fall 2025, Spring 2023, Fall 2022, Spring 2022, Fall 2021.
- 2025 Simon Marais Mathematics Competition East Division, 3rd place; [link](#)
- 43th Korean College Mathematics Competition, Gold Medal
- Mar 2021 – Present National Science & Technology Scholarship

Publications

- 2026 **Subsampling Confidence Bound for Persistent Diagram via Time-delay Embedding**
[Arxiv link](#)
Donghyun Park, Junhyun An, Taehyoung Kim, Jisu Kim.
Submitted.

Research experience

- Feb 2026 – Present **Research Internship, ‘High-Dimensional Multiple Testing’ Lab**
Mentors: Junyong Park (Seoul National University).
◦ Currently conducting research on exponential families using E-value analysis to evaluate growth rate optimal E-value.
- Apr 2025 – Jan 2026 **Research Internship, ‘Statistics and Artificial Intelligence with Topology’ Lab**
Mentors: Jisu Kim (Seoul National University).
◦ Investigated various reconstruction methods for distribution support from samples, along with the end-to-end pipeline of topological data analysis (TDA) and its applications in artificial intelligence.
- Sep 2025 – Dec 2025 **Student-Directed Education Program**
Mentors: Jisu Kim (Seoul National University).
◦ Collaborative work with two undergraduate colleagues (Taehyoung Kim and Junhyun An), Advised by Jisu Kim.
◦ Conducted research to develop a confidence bound of persistent diagrams via time-delayed embedding.

Teaching experience

- Mar 2026 – Present **Peer Tutor, Regression Analysis and Practice (Seoul National University)**
◦ Guided peers in overcoming difficulties in theoretical frameworks and practical applications.
- Dec 2025 – Jan 2026 **Mentoring Program “An Invitation to the p-adic numbers”**
◦ Designed and delivered a 4-week intensive lecture series titled “*An Invitation to the p-adic numbers*” for a cohort of 6 undergraduate peers.
◦ Key topics included analysis and algebra on p-adic numbers, Hensel’s lemma, and Gauss’s sum of three squares theorem.

Extracurricular Activities

Meari, School band of Seoul National University
King’s Gambit, Participated in regular chess matches

Skills

Programming Language

◦ Proficient at C++, Python, R

Languages & Test Scores

English: Fluent (TOEFL iBT: 5.0/6.0, GRE: Verbal 158, Quantitative 170, AWA 3.0)

Relevant Coursework

Statistics

Mathematical statistics, Regression analysis, Bayesian statistics, Data mining, Probability theory, Deep learning theory, Nonparametric statistics

Mathematics

Measure theory, Differential geometry, Morse theory, Partial differential equation, Mathematical optimization, Mathematical foundations of deep neural networks, Numerical linear algebra